

LiteTest

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LiteTest is a lightweight framework and Application Programming Interface (API) for defining, running, and reporting tests in C/C++ projects. It provides a simple core macro-driven Runner API plus a function-based Test API, fault-tolerant execution, and clear reporting. It is ideal for small to medium C projects that need reliable testing without heavy tooling and dependencies. It can be used for unit and command-line testing.

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Preface

LiteTest provides a Runner API and a Test API, each implemented with a single `.h` / `.c` pair with no external dependencies and requiring only a POSIX.1-2001 environment and a C99-compliant compiler.

For a list of other LiteTest documents and the repository layout, see the LiteTest Documentation Guide (`LiteTest_Documentation_Guide.md`).

For a glossary of terms, see the LiteTest Glossary Reference (`LiteTest_Glossary_Reference.md`).

Document Version History

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- The **Document** column records the document's version with the format `M.u` (Major, update).
- The **Runner** column records the LiteTest Runner API version current at the time this document version was published and is defined by its `LT_RUNNER_VERSION` macro.
- The **Test** column records the LiteTest Test API version current at the time this document version was published and is defined by its `LT_TEST_VERSION` macro.
- Both Runner and Test use the version format "`M.m.p`" (Major, minor, patch).
- `M` is the same for the Document, Runner, and Test versions.

The document's update version tracks updates to this document and does not correspond to a minor or patch version. `u` increments whenever this document is updated without a change to `M`, and it resets to 0 when `M` is incremented.

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1. Introduction

1.1. Key Features

- Pure C implementation
- Minimal footprint (single header + source)
- Fault detection using POSIX signals
- Optional process-isolated execution
- Parallel and concurrent test execution
- Clear pass/fail/fault reporting
- Simple API based on macros and test group functions

1.2. Quick Start

A test executable consists of your tests as C/C++ expressions/functions, and the orchestrator and test group functions you write using the LiteTest API that manages the execution.

1. Copy 'litetest_runner.h' and 'litetest_runner.c' to your current directory:

```
cp /path/to/litetest_runner.h .
cp /path/to/litetest_runner.c .
```

2. Create a file named `test_quick.c` in the same directory.

3. Copy and paste the code into `test_quick.c`:

```
#include "litetest_runner.h"

// A simple test group function.

static LT_DECLARE_GROUP(test_quick)
{
    LT_INIT_GROUP(test_quick, 1);

    int a = 2;
    int b = 2;

    // 4 test assertions.
    LT_TEST(a == b, , 0);      // Pass
    LT_TEST(a + b == 4, , 0);  // Pass
    LT_TEST(a - b == 1, , 0);  // Fail
    LT_TEST(LT_FAULT(1), , 0); // Fault

    LT_RETURN;
}

// A simple orchestrator (main) function.

LT_DECLARE_ORCHESTRATOR(main)
{
```

```

LT_INIT_ORCHESTRATOR(main, quick, 1);
LT_PARSE_ARGS(2, "quick_test_report.txt");
LT_OPEN_REPORT("Test Quick Report");

// Single test category.
LT_WRITE_RESULT(LT_GROUP(test_quick), "Quick tests");

LT_CLOSE_REPORT("Note: This report is a very simple example of using LiteTest.\n"
    "Note: Multiple test categories can be added using multiple\n"
    "    test functions.\n"
    "Note: Orchestrator (`main`) and test functions can be placed in\n"
    "    individual modules (.c files).\n"
    "Note: Parameters can be set to run tests in parallel, isolate\n"
    "    a test to a separate thread or process, etc.\n"
    "Note: The expression for LT_TEST can reference functions to\n"
    "    provide a more complex test. A non-zero result indicates\n"
    "    pass and a zero result indicates fail. If a fault occurs\n"
    "    executing the expression, it is detected and counted in\n"
    "    the report as a fault.\n"
    "Note: Larger projects can place files in a more conventional\n"
    "    layout (e.g., `include/` and `src/`, but this example keeps\n"
    "    everything in your current directory for simplification.\n"
    "Note: See README.md for LiteTest for additional API features.\n");
LT_EXIT;
}

```

4. Build the executable `test_quick` in your current directory:

```
cc -std=c99 -Wall -Wextra -o test_quick test_quick.c litetest_runner.c
```

5. Run it:

```
./test_quick
```

6. View `quick_test_report.txt` in your current directory:

```
less quick_test_report.txt    # Press 'q' to quit
```

Example of the report:

```

LiteTest Report

```

	Pass	Fail	Fault
1. Orchestrator	4		
2. Guard 1 and 2	6		
Total	10		

1.3. Requirements

Supported compilers, C standard level, and any platform notes.

1.4. Installation

How to add LiteTest to your project, including copying the header, adding the source file, or integrating via your build system.

2. LiteTest Runner Framework and API

The Runner framework (implemented using the Runner API) executes all test expressions (that implement the tests), aggregates results, and produces the report. This is the framework used by your test executable.

The Runner API provides typedefs, enums, macros, and functions for writing tests. Developers use this API to express expected behavior, group related tests, and define optional setup/teardown logic.

2.1. What the Runner Framework and API Does

The Runner framework coordinates test execution from the orchestrator down to individual test groups. It initializes the run, applies include and concurrency settings, executes the requested groups, captures pass/fail/fault results, and writes the final report.

The Runner API provides the macros and helper types used to define the orchestrator, declare test groups, control execution flow, and handle fault detection and reporting.

3. Running Tests

How to invoke the runner from `main()`, including optional parameters such as output paths or configuration flags.

4. Report Generation

How LiteTest collects pass/fail information and formats it for output.

5. Customization

Notes that you can build your own runner logic if you need custom ordering, filtering, or integration behavior.

6. Test Report

Describes the structure of the test report, failures, optional details, and how to consume it in tooling or CI.

7. Advanced Features and Topics

Filtering tests and custom report writers are available in the User Guide.

8. Test Expressions

How to write a test expression for `LT_TEST(expr)` and how expressions are evaluated.

9. Setup and Teardown

Optional per-group or per-test initialization and cleanup helpers.

10. Test Groups

How to organize related tests into groups for readability and logical structure.

11. Concurrent Tests

How to mark tests as concurrent and how LiteTest handles parallel execution safely.

12. Isolation

Isolation controls how much separation LiteTest uses when running tests.

- **Same-thread execution** is the default and has the lowest overhead.
- **Thread-isolated execution** keeps tests separated without switching to a separate process.
- **Process-isolated execution** provides the strongest containment for faults, crashes, aborts, and other disruptive failures.

Use the lightest mode that still gives you the safety you need for the code under test.

13. LiteTest Test API

The Test API provides macros and functions for writing tests. Developers use this API to express expected behavior, group related tests, and define optional setup/teardown logic.

13.1 File Functions

These helpers create, inspect, read, and modify files and directories.

13.1.1 Filesystem Predicate Functions These helpers answer simple questions about a path or file.

- `lt_exists`
- `lt_is_file`
- `lt_is_directory`
- `lt_get_size`
- `lt_file_age`
- `lt_path_has_extension`

Example:

```
if (lt_exists(path) == 1 && lt_is_file(path) == 1) {  
    LT_ASSERT(lt_path_has_extension(path, ".txt") == 1, 0);  
}
```

13.1.2 File and Directory Operation Functions These helpers create, read, write, and transform file-system content.

- `lt_copy_file`
- `lt_make_dirs`
- `lt_path_join`
- `lt_read_file`
- `lt_read_file_with_limit`
- `lt_read_file_into`
- `lt_remove_tree`
- `lt_stat_check`
- `lt_touch`
- `lt_write_file`
- `lt_append_file`
- `lt_rename_file`
- `lt_symlink`

Example:

```
char output_path[1024];
```

```
LT_ASSERT(lt_path_join(root, "tmp/output.txt", output_path, sizeof output_path) == LT_OK  
LT_ASSERT(lt_write_file(output_path, data, length) == LT_OK, 0);
```

13.1.3 Temporary and Cleanup Helpers These helpers create temporary resources and clean them up when a test ends.

- `lt_make_temp_dir`
- `lt_make_temp_file`
- `lt_cleanup_register`
- `lt_temp_file_auto`

Example:

```
static void cleanup_temp_file(void *context)
{
    (void)lt_remove_tree((const char *)context);
}
```

```
char temp_path[1024];
```

```
LT_ASSERT(lt_make_temp_file(".txt", temp_path, sizeof temp_path) == LT_OK, 0);
LT_ASSERT(lt_cleanup_register(cleanup_temp_file, temp_path) == LT_OK, 0);
```

13.2 Compare Functions

These helpers compare files, metadata, JSON, text, and patterns. Most return `LT_OK` / `LT_MISMATCH` or 1 / 0 depending on the helper family.

13.2.1 Path, File, and Directory Comparisons These helpers compare file contents, paths, and directory trees.

- `lt_compare_dirs`
- `lt_compare_files`
- `lt_compare_file_to_path`
- `lt_compare_paths`
- `lt_compare_path_to_file`
- `lt_compare_paths_with_options`
- `lt_compare_paths_ignoring_patterns`
- `lt_compare_paths_masking_fields`
- `lt_compare_paths_masking_ranges`
- `lt_compare_file_lines`

Example:

```
const lt_path_compare_options_t options = LT_PATH_COMPARE_OPTIONS_INIT;
```

```
LT_ASSERT(lt_compare_paths_with_options(expected_path, actual_path, &options) == LT_OK,
```

13.2.2 Metadata and Binary Comparisons These helpers compare file metadata and low-level binary content.

- `lt_compare_path_metadata`
- `lt_hexdump_diff`

- `lt_compare_memory_detail`

Example:

```
const lt_path_metadata_compare_options_t options = LT_PATH_METADATA_COMPARE_OPTIONS_INIT;

LT_ASSERT(lt_compare_path_metadata(left_path, right_path, &options) == LT_OK, 0);
```

13.2.3 JSON Helpers These helpers compare or query JSON documents.

- `lt_compare_json`
- `lt_compare_json_with_limit`
- `lt_json_extract`
- `lt_json_has_path`

Example:

```
LT_ASSERT(lt_json_has_path(json_text, "response.items[0].id") == 1, 0);
```

13.2.4 Text and Pattern Helpers These helpers compare or match text data.

- `lt_compare_text_normalized`
- `lt_match`
- `lt_match_regex`
- `lt_string_contains`
- `lt_string_starts_with`
- `lt_string_ends_with`

Example:

```
LT_ASSERT(lt_string_contains(message, "success", NULL) == 1, 0);
```